

Monomial $f(q) = a \cdot q^8$ Report

Polynomial

$$f(q) = aq^8$$

Naive

$$g_{\text{naive}}(X) = X^8 a$$

$$\mathbb{E}[g_{\text{naive}}(X)] = 40320a \frac{\Delta^8}{\epsilon} + 20160a \frac{\Delta^6}{\epsilon} q^2 + 1680a \frac{\Delta^4}{\epsilon} q^4 + 56a \frac{\Delta^2}{\epsilon} q^6 + aq^8$$

$$\begin{aligned} \text{Var}(g_{\text{naive}}(X)) &= 20921164185600a^2 \frac{\Delta^{16}}{\epsilon} + 10459769241600a^2 \frac{\Delta^{14}}{\epsilon} q^2 \\ &\quad + 871241011200a^2 \frac{\Delta^{12}}{\epsilon} q^4 + 28987176960a^2 \frac{\Delta^{10}}{\epsilon} q^6 \\ &\quad + 513757440a^2 \frac{\Delta^8}{\epsilon} q^8 + 5537280a^2 \frac{\Delta^6}{\epsilon} q^{10} \\ &\quad + 37184a^2 \frac{\Delta^4}{\epsilon} q^{12} + 128a^2 \frac{\Delta^2}{\epsilon} q^{14} \end{aligned}$$

$$\begin{aligned} \text{MSE}(g_{\text{naive}}) &= 20922789888000a^2 \frac{\Delta^{16}}{\epsilon} + 10461394944000a^2 \frac{\Delta^{14}}{\epsilon} q^2 \\ &\quad + 871782912000a^2 \frac{\Delta^{12}}{\epsilon} q^4 + 29059430400a^2 \frac{\Delta^{10}}{\epsilon} q^6 \\ &\quad + 518837760a^2 \frac{\Delta^8}{\epsilon} q^8 + 5725440a^2 \frac{\Delta^6}{\epsilon} q^{10} \\ &\quad + 40320a^2 \frac{\Delta^4}{\epsilon} q^{12} + 128a^2 \frac{\Delta^2}{\epsilon} q^{14} \end{aligned}$$

$$\text{Bias}(g_{\text{naive}}) = 40320a \frac{\Delta^8}{\epsilon} + 20160a \frac{\Delta^6}{\epsilon} q^2 + 1680a \frac{\Delta^4}{\epsilon} q^4 + 56a \frac{\Delta^2}{\epsilon} q^6$$

Unbiased

$$g_{\text{unb}}(X) = X^8 a - 56X^6 a \frac{\Delta^2}{\epsilon}$$

$$\mathbb{E}[g_{\text{unb}}(X)] = aq^8$$

$$\begin{aligned}
\text{Var}(g_{\text{unb}}(X)) &= 12660970291200a^2 \frac{\Delta^{16}}{\epsilon} + 6330485145600a^2 \frac{\Delta^{14}}{\epsilon} q^2 \\
&\quad + 527540428800a^2 \frac{\Delta^{12}}{\epsilon} q^4 + 17584680960a^2 \frac{\Delta^{10}}{\epsilon} q^6 \\
&\quad + 314012160a^2 \frac{\Delta^8}{\epsilon} q^8 + 3489024a^2 \frac{\Delta^6}{\epsilon} q^{10} \\
&\quad + 26432a^2 \frac{\Delta^4}{\epsilon} q^{12} + 128a^2 \frac{\Delta^2}{\epsilon} q^{14}
\end{aligned}$$

$$\begin{aligned}
\text{MSE}(g_{\text{unb}}) &= 12660970291200a^2 \frac{\Delta^{16}}{\epsilon} + 6330485145600a^2 \frac{\Delta^{14}}{\epsilon} q^2 \\
&\quad + 527540428800a^2 \frac{\Delta^{12}}{\epsilon} q^4 + 17584680960a^2 \frac{\Delta^{10}}{\epsilon} q^6 \\
&\quad + 314012160a^2 \frac{\Delta^8}{\epsilon} q^8 + 3489024a^2 \frac{\Delta^6}{\epsilon} q^{10} \\
&\quad + 26432a^2 \frac{\Delta^4}{\epsilon} q^{12} + 128a^2 \frac{\Delta^2}{\epsilon} q^{14}
\end{aligned}$$

Comparison

$$\text{mean gap} = -40320a \frac{\Delta^8}{\epsilon} - 20160a \frac{\Delta^6}{\epsilon} q^2 - 1680a \frac{\Delta^4}{\epsilon} q^4 - 56a \frac{\Delta^2}{\epsilon} q^6$$

$$\begin{aligned}
\text{variance gap} &= -8260193894400a^2 \frac{\Delta^{16}}{\epsilon} - 4129284096000a^2 \frac{\Delta^{14}}{\epsilon} q^2 \\
&\quad - 343700582400a^2 \frac{\Delta^{12}}{\epsilon} q^4 - 11402496000a^2 \frac{\Delta^{10}}{\epsilon} q^6 \\
&\quad - 199745280a^2 \frac{\Delta^8}{\epsilon} q^8 - 2048256a^2 \frac{\Delta^6}{\epsilon} q^{10} - 10752a^2 \frac{\Delta^4}{\epsilon} q^{12}
\end{aligned}$$

$$\begin{aligned}
\text{MSE gap} &= -8261819596800a^2 \frac{\Delta^{16}}{\epsilon} - 4130909798400a^2 \frac{\Delta^{14}}{\epsilon} q^2 \\
&\quad - 344242483200a^2 \frac{\Delta^{12}}{\epsilon} q^4 - 11474749440a^2 \frac{\Delta^{10}}{\epsilon} q^6 \\
&\quad - 204825600a^2 \frac{\Delta^8}{\epsilon} q^8 - 2236416a^2 \frac{\Delta^6}{\epsilon} q^{10} - 13888a^2 \frac{\Delta^4}{\epsilon} q^{12}
\end{aligned}$$

$$\begin{aligned}
\text{Bias}(g_{\text{naive}})^2 &= 1625702400a^2 \frac{\Delta^{16}}{\epsilon} + 1625702400a^2 \frac{\Delta^{14}}{\epsilon} q^2 \\
&\quad + 541900800a^2 \frac{\Delta^{12}}{\epsilon} q^4 + 72253440a^2 \frac{\Delta^{10}}{\epsilon} q^6 \\
&\quad + 5080320a^2 \frac{\Delta^8}{\epsilon} q^8 + 188160a^2 \frac{\Delta^6}{\epsilon} q^{10} + 3136a^2 \frac{\Delta^4}{\epsilon} q^{12}
\end{aligned}$$